MTH636 – Game Theory

Lecture 29: Strategic Form $\leftrightarrow$ Extensive Form & Incomplete Information Games

1 From Strategic Form to Extensive Form

1.1 Motivation

Suppose you have a strategic form game G.

- **Question 1:** Can you get an equivalent extensive form game?

$$G \longrightarrow \Gamma \longrightarrow G \quad \text{(equivalence)}$$

- **Question 2:** If it exists, whether such an extensive form is unique or not?

1.2 Answer to Question 1

Answer to Question 1: Yes. Given

$$G = \left\langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \right\rangle,$$

assume that $S_i = \{s_i^1, \ldots, s_i^{m_i}\}$ for each i.

We construct an extensive form game Γ as follows. The tree has the structure:

- The root belongs to Player I. She has m_1 actions: $s_1^1, s_1^2, \ldots, s_1^{m_1}$.
- Each of Player I's m_1 children belongs to Player II (all in a single information set – Player II cannot observe Player I's move). Player II has m_2 actions: $s_2^1, s_2^2, \ldots, s_2^{m_2}$.
- And so on for Player III, ..., Player n, each player having a single information set (so they cannot observe any previous moves).
- Every leaf node corresponds to a unique strategy profile in the strategic form game. The payoffs at that leaf are the utilities would u_i of the corresponding profile.

Note

Every leaf node corresponds to a unique strategy profile in the strategic form game.

It is easy to see that the corresponding strategic form game of the above extensive form game is G only.

1.3 Answer to Question 2 (Uniqueness)

If you change the ordering of the players in the extensive form game, you will get another equivalent form.

Example

Consider the following 2×4 strategic form game between Player 1 (rows L, R) and Player 2 (columns rt, rb, ℓt, ℓb):

$1 \setminus 2$	rt	rb	ℓt	ℓb
L	$(2,-1)$	$(2,-1)$	$(2,0)$	$(2,0)$
R	$(2,1)$	$(1,0)$	$(2,1)$	$(1,0)$

This strategic form game can be represented as an extensive form game in (at least) two ways:

Extensive form with Player 1 moving first:

- Player I moves first: L or R.
- After L: Player II is in an information set and chooses r or ℓ (each leading to further choices).
- After R: Player II is in an information set and chooses r or ℓ, etc.
- Player II's strategy space consists of combinations $\{rt, rb, \ell t, \ell b\}$.

The resulting extensive form tree (Player I at root, Player II at second level in a single information set):

- Root: Player I chooses L or R.
- Left branch (L): Player II (single info set) chooses r or ℓ.

  r: leads to leaf with payoffs $(2,-1)$.

  ℓ: leads to leaf with payoffs $(2,0)$.
- Right branch (R): Player II (same info set? or separate) chooses t or b (or r and ℓ leading to t/b).

Alternative extensive form (Player II depicted first at the single-info-set level): A cleaner representation places Player I at the root choosing L or R, then Player II at two separate nodes (each a subgame given I's move), both enclosed in a single information set (since Player II does not observe Player I's choice). Terminal payoffs match the matrix entries.

Extensive form tree (Player I first):

Player I: L or R

$\Downarrow$

Player II (information set): n (right) or ℓ (left)

$\Downarrow$

Leaves: $(2,-1)$, $(2,0)$ after L; $(2,1)$, $(1,0)$ after R with r, ℓ respectively.

Note

The extensive form representation of a strategic form game is **not unique**. Different orderings of the players yield different (but equivalent) extensive form games.

2 Incomplete Information Games

2.1 Motivation

This model is due to **Harsanyi (1967)**.

Motivating Example. Fix 13 cards. We can calculate the probabilities of the cards the other players are having:

$$\binom{52}{13}\binom{39}{13}\binom{26}{13}\binom{13}{13} \quad \text{or equivalently} \quad \binom{39}{13}\binom{26}{13}.$$

Since each player's hand is drawn from the remaining cards, the type of each player (which 13 cards they hold) is private information – other players only know the distribution.

2.2 Definition

Definition: Incomplete Information Game

An incomplete information game is a tuple:

$$\left\langle N,\ (T_i)_{i\in N},\ p,\ S,\ (s_A)_{A\in S\times T_i} \right\rangle$$

where:

- N: a finite set of players.
- T_i: a finite set of **types** for player i. The set of type vectors (type profiles) is denoted by $T = \bigtimes_{i\in N} T_i$.

  (In the card example, $|T_i| = \binom{52}{13}$; and $|T| = \left[\binom{52}{13}\right]^4$.)
- p: a probability distribution over T. We assume that

$$p(t_i) \;=\; \sum_{t_{-i}\in T_{-i}} p(t_i, t_{-i}) \;>\; 0 \qquad \forall\, i \in N,\ \ \forall\, t_i \in T_i.$$

  (Every type of every player is possible.)
- S: a set of **state games**. Every state game $s \in S$ is itself a vector:

$$s = \left\langle N,\ (A_i(t_i))_{i\in N, t_i\in T_i},\ (u_i)_{i\in N} \right\rangle$$

  where A_i is a non-empty set of actions available to player i, and $u_i : X \times A_i \to \mathbb{R}$ (where the product is taken over $i \in N$) is the utility function of player i.
- s_A: an element in S – the state game corresponding to a type profile.

Note

It is important to note that player i's action depends on **his own type only**. Specifically, s_A is an element in S where:

$$s_A = \left\langle N,\ (A_i(t_i))_{i\in N, t_i\in T_i},\ (u_i(t))_{i\in N} \right\rangle.$$

Player i's action is chosen from $A_i(t_i)$, which depends on t_i (player i's own type) only.